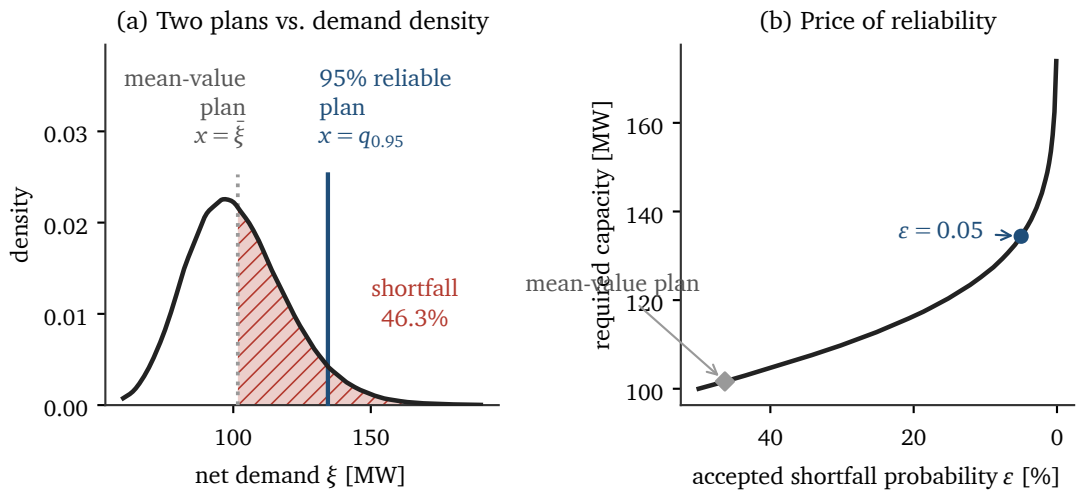
(a) Two plans vs. demand density
mean-value plan
$x = \bar{\xi}$
95% reliable plan
$x = q_{0.95}$
shortfall
46.3%
density
0.00
0.01
0.02
0.03
100
150
net demand ξ [MW]
(b) Price of reliability
mean-value plan
$\varepsilon = 0.05$
required capacity [MW]
100
120
140
160
40
20
0
accepted shortfall probability ε [%]